

Target board: Artix A7-100T

Worst negative slack: 0.086ns

Synthesis strategy: default

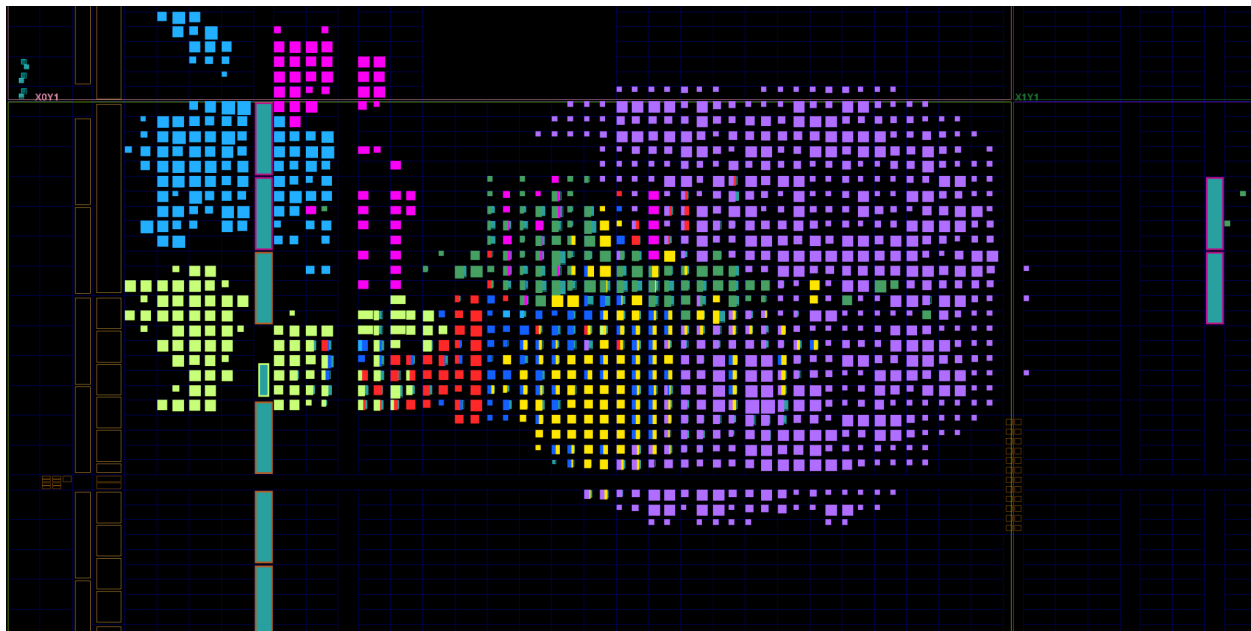
Target FPGA: XC7A100TCSG324-1

Worst hold slack: 0.056ns

Implementation strategy: default

Target clock: 100 MHz

Floor Plan



Utilization Report

▼ riscv_soc	2324	2219	293	140	1098	2223	101	8.5	4	1
▼ u_core (riscv_core)	1780	1642	273	134	918	1701	79	0	0	0
u_branch_predictor (branch_predictor)	272	64	16	6	81	193	79	0	0	0
u_decoder (decoder)	502	134	0	0	196	502	0	0	0	0
u_execute (execute)	73	111	0	0	117	73	0	0	0	0
u_fetch (fetch)	133	131	1	0	95	133	0	0	0	0
u_memory (memory)	114	109	0	0	69	114	0	0	0	0
u_reg_file (reg_file)	683	992	256	128	573	683	0	0	0	0
u_l1_dcache (l1_dcache)	295	204	16	6	99	273	22	0.5	0	0
u_uart_packet_parser (uart_packet_parser)	95	86	0	0	35	95	0	0	0	0
u_uart_packet_serializer (uart_packet_serializer)	37	88	3	0	27	37	0	0	0	0
u_uart_program_loader (uart_program_loader)	58	122	0	0	47	58	0	0	0	0

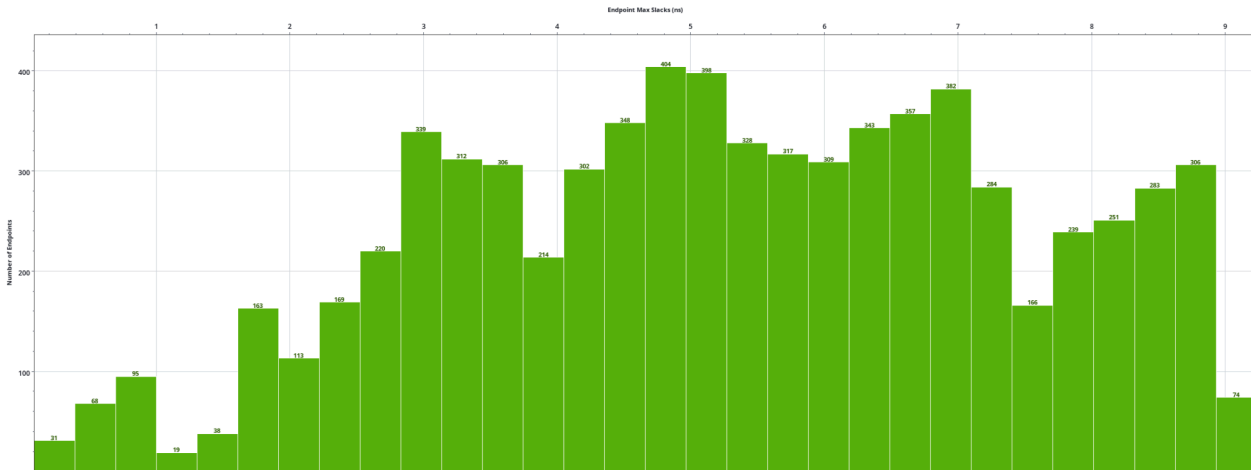
Setup Time Critical Paths

Intra-Clock Paths - clk_100 - Setup										
Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	
Path 1	0.086	11	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...esult_reg[25]/D	9.781	1.882	7.899	10.000	
Path 2	0.089	11	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...esult_reg[15]/D	9.773	1.882	7.891	10.000	
Path 3	0.127	11	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[29]/D	9.756	1.882	7.874	10.000	
Path 4	0.142	11	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[9]/D	9.710	1.882	7.828	10.000	
Path 5	0.147	14	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...result_reg[0]/D	9.800	2.629	7.171	10.000	
Path 6	0.155	16	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...esult_reg[22]/D	9.687	3.175	6.512	10.000	
Path 7	0.160	14	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolve_taken_q_reg/D	9.790	2.811	6.979	10.000	
Path 8	0.162	11	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...esult_reg[31]/D	9.703	1.882	7.821	10.000	
Path 9	0.169	11	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[6]/D	9.690	1.882	7.808	10.000	
Path 10	0.179	11	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[10]/D	9.674	1.882	7.792	10.000	
Path 11	0.194	18	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[30]/D	9.648	3.391	6.257	10.000	
Path 12	0.200	16	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[22]/D	9.682	3.175	6.507	10.000	
Path 13	0.212	11	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...esult_reg[18]/D	9.630	1.882	7.748	10.000	
Path 14	0.221	11	64	u_core/u_execute/...result_reg[27]/C	u_core/bp_resolv...rget_q_reg[24]/D	9.621	1.882	7.739	10.000	
Path 15	0.221	11	64	u_core/u_execute/...result_reg[27]/C	u_core/u_execute...result_reg[8]/D	9.638	1.882	7.756	10.000	

Hold Time Critical Paths

Intra-Clock Paths - clk_100 - Hold											
Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock
Path 16	0.056	0	1	u_uart_program_l...r_data_reg[29]/C	u_imem/memory_reg_3/DIADI[2]	0.270	0.164	0.106	0.000	clk_100	clk_100
Path 17	0.077	0	2	u_l1_dcache/req_tag_reg[15]/C	u_l1_dcache/tag...0_63_15_15/SP/I	0.260	0.141	0.119	0.000	clk_100	clk_100
Path 18	0.080	0	1	u_uart_program_l...r_data_reg[7]/C	u_imem/memory_reg_0/DIADI[7]	0.240	0.128	0.112	0.000	clk_100	clk_100
Path 19	0.085	0	2	u_l1_dcache/req_tag_reg[5]/C	u_l1_dcache/tag...g_0_63_5_5/SP/I	0.268	0.141	0.127	0.000	clk_100	clk_100
Path 20	0.088	0	116	u_core/bp_resolve_pc_q_reg[4]/C	u_core/u_branc...6_8/RAMA/WADR2	0.380	0.141	0.239	0.000	clk_100	clk_100
Path 21	0.088	0	116	u_core/bp_resolve_pc_q_reg[4]/C	u_core/u_branc...6_8/RAMB/WADR2	0.380	0.141	0.239	0.000	clk_100	clk_100
Path 22	0.088	0	116	u_core/bp_resolve_pc_q_reg[4]/C	u_core/u_branc...6_8/RAMC/WADR2	0.380	0.141	0.239	0.000	clk_100	clk_100
Path 23	0.088	0	116	u_core/bp_resolve_pc_q_reg[4]/C	u_core/u_branc...6_8/RAMD/WADR2	0.380	0.141	0.239	0.000	clk_100	clk_100
Path 24	0.093	0	2	u_l1_dcache/req_tag_reg[12]/C	u_l1_dcache/tag...0_63_12_12/SP/I	0.273	0.141	0.132	0.000	clk_100	clk_100
Path 25	0.094	0	1	u_uart_program_l...r_data_reg[27]/C	u_imem/memory_reg_3/DIADI[0]	0.309	0.164	0.145	0.000	clk_100	clk_100
Path 26	0.095	0	43	u_l1_dcache/req_index_reg[4]/C	u_l1_dcache/tag...63_8_8/SP/ADR4	0.331	0.141	0.190	0.000	clk_100	clk_100
Path 27	0.095	0	43	u_l1_dcache/req_index_reg[4]/C	u_l1_dcache/tag...63_9_9/SP/ADR4	0.331	0.141	0.190	0.000	clk_100	clk_100
Path 28	0.096	0	1	u_uart_program_l...r_data_reg[5]/C	u_imem/memory_reg_0/DIADI[5]	0.309	0.141	0.168	0.000	clk_100	clk_100

Slack Time Histogram

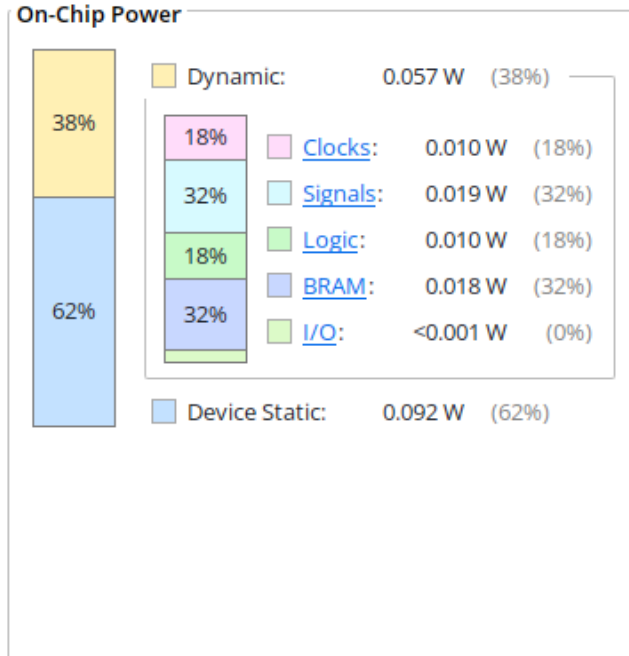


Power Utilization Report

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	0.149 W
Design Power Budget:	Not Specified
Process:	typical
Power Budget Margin:	N/A
Junction Temperature:	25.7°C
Thermal Margin	59.3°C (12.9 W)
Ambient Temperature	25.0 °C
Effective θ_{JA}	4.6°C/W
Power supplied to off-chip devices	0 W
Confidence level	Medium

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

[illegible]